

Patent Keyword Analysis Report

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Detected language: en

Warning: The displayed route is enforced as technical concept → catalog-backed IPC/CPC area → linked FI subdivision → FI-scoped F-term theme/aspect. It is search guidance, not an official classification determination; confirm the current hierarchy and scope in J-PlatPat before relying on it. Each displayed keyword passed the concept-rationale gate: a named technical-concept facet, a server-derived concept basis, and an exact input-text excerpt are required before the result can consume a credit.

AI-derived common technical concept	Shared by all keywords
Object/system	An automotive water-submersion response system that detects partial vehicle submersion and automatically adjusts a vehicle feature.
Purpose or problem	Automatically responding when a vehicle is at least partially submerged in water.
Application/use	Vehicle safety and emergency response during water immersion.
Components	vehicle, sensor, sensor data, electronic control unit (ECU), adjustable vehicle feature
Component relationships	The sensor detects sensor data indicating whether the vehicle is at least partially submerged in water., The ECU is connected to the sensor and determines partial submersion based on the sensor data., The ECU adjusts at least one vehicle feature in response to the submersion determination.
Material/composition	—
Manufacturing or processing steps	—
Operation	The sensor detects water-submersion conditions, the ECU determines whether partial submersion exists, and the ECU automatically adjusts a vehicle feature.
Control means	electronic control unit (ECU) responsive to sensor data
Controlled variable	at least one vehicle feature
Operating conditions	The vehicle is at least partially submerged in water., The ECU operates based on sensor data indicating the submersion condition.
Technical effect	Provides an automatic vehicle response to detected partial water submersion by changing at least one vehicle feature.
Context terms	automotive safety, water immersion, emergency vehicle response, condition detection, feature adjustment
Search phrases	vehicle sensor detecting partial submersion in water, electronic control unit responsive to vehicle water-submersion sensor data, automatically adjusting a vehicle feature after partial water immersion, vehicle feature control based on detected submersion

Keyword 1	vehicle
Normalized term	vehicle
Synonyms	automobile, motor vehicle, automotive vehicle
IPC	—
CPC	—
FI	—
F-term	—

Keyword 1	vehicle
Occurrences	5
Rank	1
Confidence	low
Reason	“vehicle” is grounded in this disclosed technical application: “partial submersion of a vehicle in water” . This specific role makes the term useful for retrieving patents with the same disclosed technical application.

Keyword 2	sensor
Normalized term	submersion sensor
Synonyms	detection sensor, water-detection sensor, condition sensor
IPC	—
CPC	B60N 2230/30
FI	—
F-term	—
Occurrences	3
Rank	2
Confidence	medium
Reason	“sensor” is grounded in this disclosed technical operation: “a sensor configured to detect sensor data indicating whether the vehicle is at least partially submerged in water” . This specific role makes the term useful for retrieving patents with the same disclosed technical operation.

Keyword 3	electronic control unit
Normalized term	electronic control unit
Synonyms	ECU, vehicle control unit, automotive controller
IPC	—
CPC	—
FI	—
F-term	—
Occurrences	3
Rank	3
Confidence	low
Reason	“electronic control unit” is grounded in this disclosed control mechanism: “The system includes an electronic control unit (ECU) connected to the sensor.” . This specific role makes the term useful for retrieving patents with the same disclosed control mechanism.

Keyword 4	sensor data
Normalized term	submersion sensor data
Synonyms	detection data, vehicle condition data, water-immersion data
IPC	—
CPC	G06Q 40/08221, B60N 2230/30
FI	—
F-term	—
Occurrences	2
Rank	4
Confidence	medium
Reason	“sensor data” is grounded in this disclosed technical operation: “detect sensor data indicating whether the vehicle is at least partially submerged in water” . This specific role makes the term useful for retrieving patents with the same disclosed technical operation.

Search Query Strategies	Database-specific starting points
AI query review - accepted	AI reviewed the generated search-query options, and server validation retained only approved keyword options and database-verified CPC candidates.
Recommended strategy	Query 2 - Balanced. Query 1 is broader; Query 3 is more selective.

Google Patents	Google Patents Advanced Search
Platform note	Uses Boolean keyword groups plus database-verified CPC codes when available.
Query 1 - Broad discovery	High-recall text search for terminology, neighboring concepts, and seed documents. ("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor")
Query 2 - Balanced (recommended)	Recommended starting point: combines the principal concepts with verified CPC scope when available. ("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor") AND ("electronic control unit" OR "ECU" OR "vehicle control unit") AND (CPC=B60N2230/30 OR CPC=G06Q40/08221)
Query 3 - Precision	Uses additional concept groups when available; otherwise narrows synonym alternatives while retaining verified CPC scope. ("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor") AND ("electronic control unit" OR "ECU" OR "vehicle control unit") AND ("submersion sensor data" OR "sensor data" OR "detection data") AND (CPC=B60N2230/30 OR CPC=G06Q40/08221)

PATENTSCOPE		PATENTSCOPE Advanced Search	
Platform note		Targets EN_ALLTXT and adds CPC (or IPC fallback) classification syntax when verified codes are available.	
Query 1 - Broad discovery		High-recall full-text search using the strongest concept groups. EN_ALLTXT:(("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor"))	
Query 2 - Balanced (recommended)		Recommended starting point: full-text concepts plus a verified classification condition when available. EN_ALLTXT:(("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor") AND ("electronic control unit" OR "ECU" OR "vehicle control unit")) AND CPC:(B60N2230/30 OR G06Q40/08221)	
Query 3 - Precision		Uses additional concept groups when available; otherwise narrows synonym alternatives while keeping the verified classification condition. EN_ALLTXT:(("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor") AND ("electronic control unit" OR "ECU" OR "vehicle control unit") AND ("submersion sensor data" OR "sensor data" OR "detection data")) AND CPC:(B60N2230/30 OR G06Q40/08221)	
USPTO Patent Public Search		Patent Public Search Advanced Search	
Platform note		Uses Boolean text terms and the .cpc. field code with spaces removed from CPC symbols.	
Query 1 - Broad discovery		High-recall Boolean text search for U.S. patents and published applications. ("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor")	
Query 2 - Balanced (recommended)		Recommended starting point: principal text concepts plus verified CPC scope when available. ("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor") AND ("electronic control unit" OR "ECU" OR "vehicle control unit") AND (B60N2230/30.cpc. OR G06Q40/08221.cpc.)	
Query 3 - Precision		Uses additional concept groups when available; otherwise narrows synonym alternatives while retaining the CPC field restriction. ("vehicle" OR "automobile" OR "motor vehicle") AND ("submersion sensor" OR "sensor" OR "detection sensor") AND ("electronic control unit" OR "ECU" OR "vehicle control unit") AND ("submersion sensor data" OR "sensor data" OR "detection data") AND (B60N2230/30.cpc. OR G06Q40/08221.cpc.)	
J-PlatPat		Patent/Utility Model Search - Logical Expression Input	
Platform note		Paste each expression into J-PlatPat's Logical Expression Input. The generator uses * for AND, + for OR, /TX for Full text, verified /FI when available (otherwise /IP for IPC), and /FT for verified F-term refinement.	
Query 1 - Broad discovery		High-recall J-PlatPat full-text logical expression without a classification restriction. [(vehicle+automobile+'motor vehicle')/TX]*(['submersion sensor'+sensor+'detection sensor')/TX]	
Query 2 - Balanced (recommended)		Recommended starting point: principal full-text concepts combined with verified FI scope, or IPC when verified FI is unavailable. [(vehicle+automobile+'motor vehicle')/TX]*(['submersion sensor'+sensor+'detection sensor')/TX]*(['electronic control unit'+ECU+'vehicle control unit')/TX]	
Query 3 - Precision		Uses additional full-text concept groups when available and adds verified F-term refinement when available. [(vehicle+automobile+'motor vehicle')/TX]*(['submersion sensor'+sensor+'detection sensor')/TX]*(['electronic control unit'+ECU+'vehicle control unit')/TX]*(['submersion sensor data'+sensor data+'detection data')/TX]	

These database-specific queries are search starting points. Verify the current syntax and classification scope in the target system, review the retrieved results, and refine the strategy before relying on it.